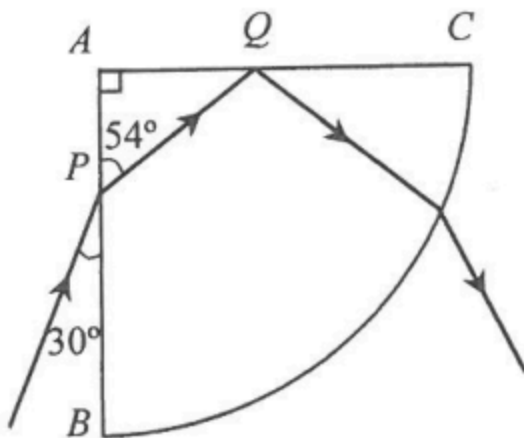
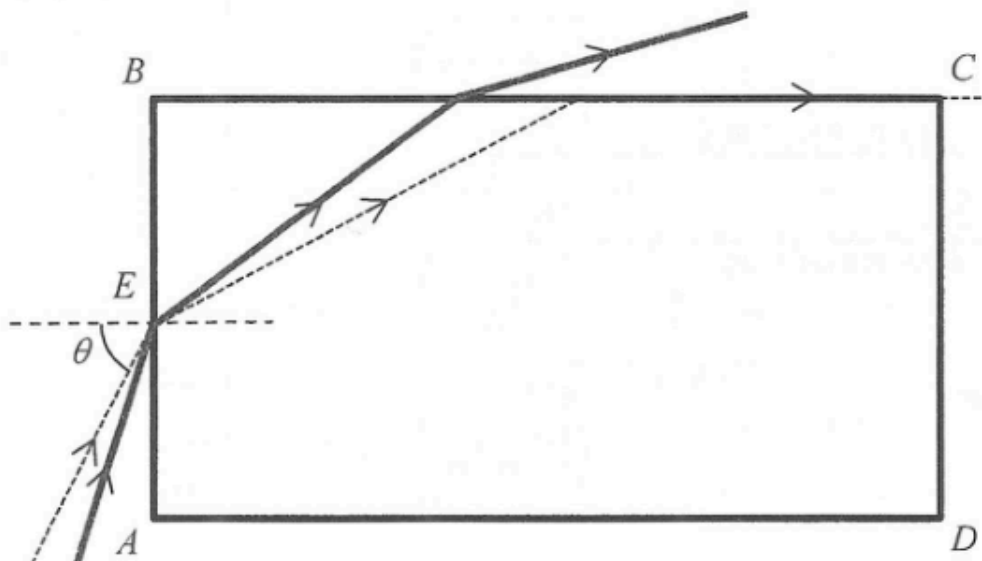
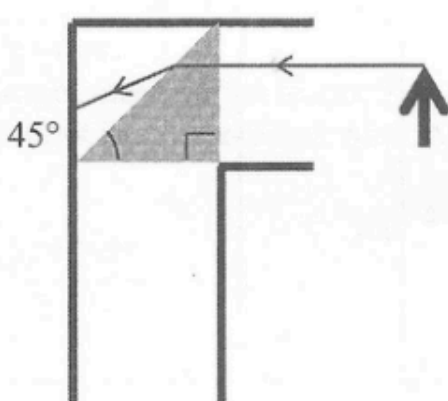
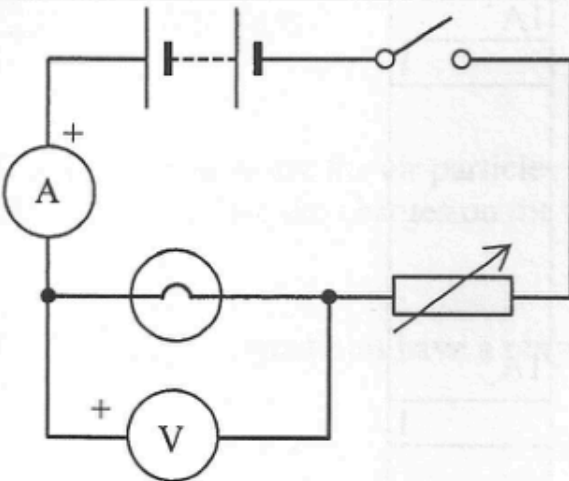
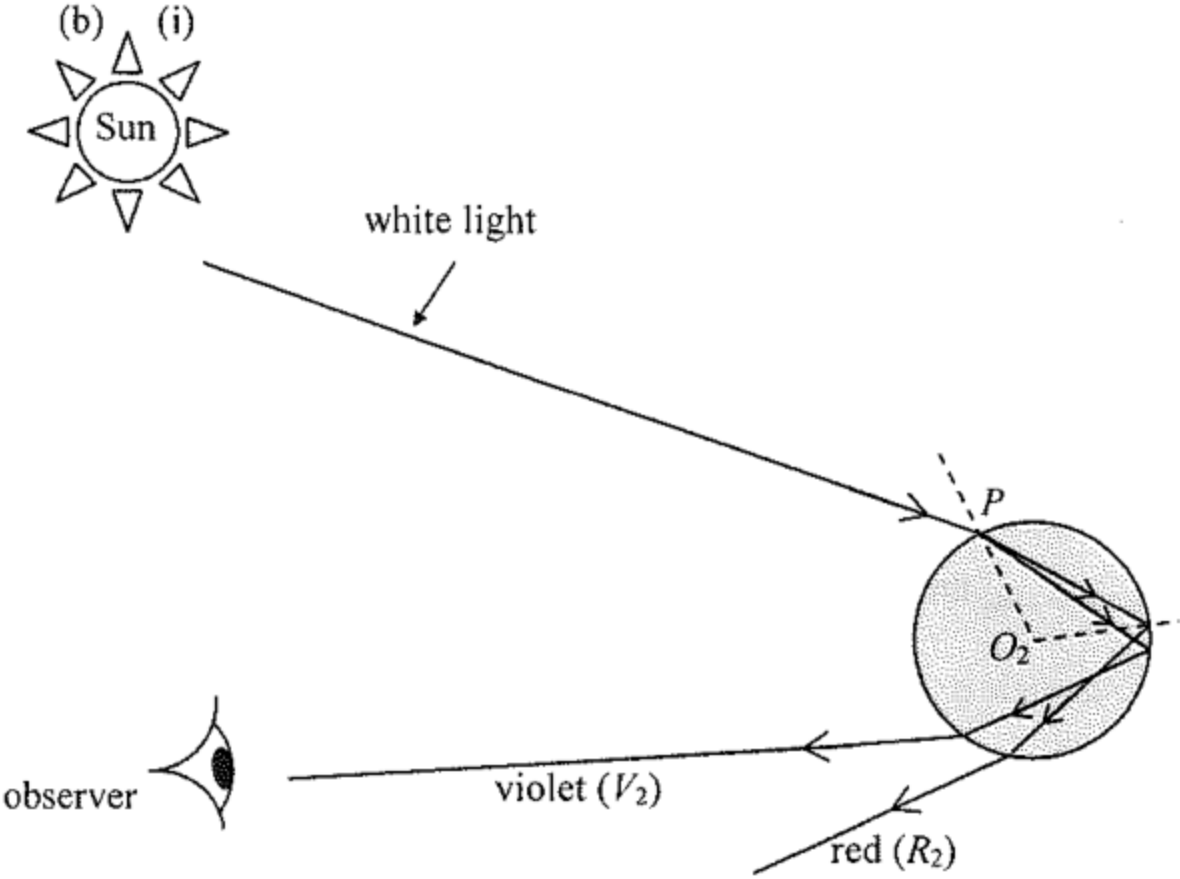


Solution	Marks	Remarks
5. (a) $n = \frac{\sin i}{\sin r}$ (or $n_1 \sin \theta_1 = n_2 \sin \theta_2$) $= \frac{\sin(90^\circ - 30^\circ)}{\sin(90^\circ - 54^\circ)} = \frac{\sin 60^\circ}{\sin 36^\circ}$ $= 1.47$	1M 1A 2	Accept 1.47 to 1.50
(b) $\sin c = \frac{1}{n} = \frac{1}{1.47}$ $c = 42.7^\circ$ (for $n = 1.50$, $c = 41.8^\circ$) Because incident angle at face AC ($= 54^\circ$) $> c$ ($= 42.7^\circ$).	1M 1M 2	
(c)  Reflected ray correct, $i = r$ Emergent ray away from normal.	1A 1A 2	
(d) A spectrum is seen. Or splitting into different colours occurs.	1A 1A 1	

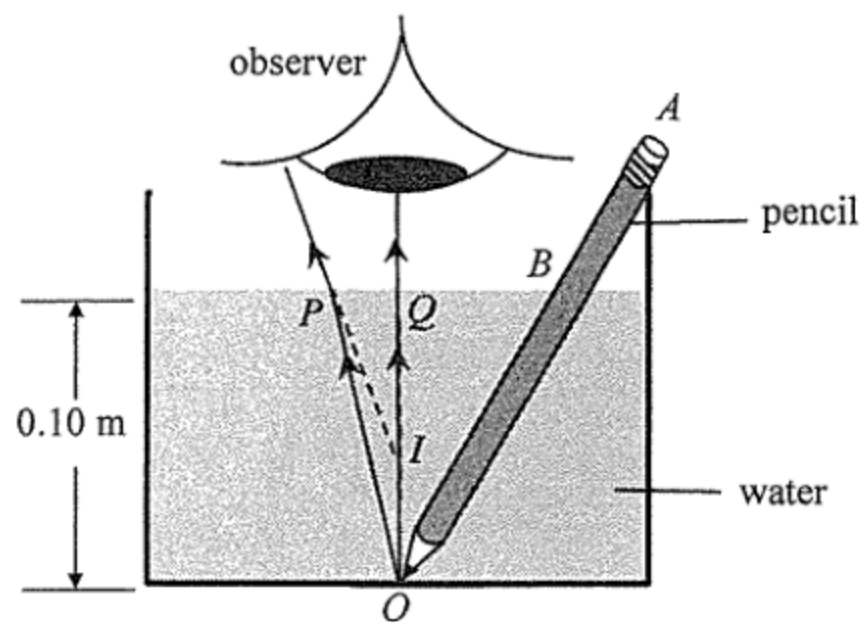
7.	(a)	(i)	At the critical angle c ,			
			$\frac{\sin 90^\circ}{\sin c} = n$		1M	
			$\frac{1}{\sin c} = 1.36$		1A	
			$c = 47.3^\circ$			2
		(ii)	Angle of refraction at $E = 90^\circ - 47.33^\circ = 42.67^\circ$		1M	
			By Snell's law			
			$\frac{\sin \theta}{\sin 42.67^\circ} = 1.36$		1M	
			$\theta = 67.2^\circ$		1A	
						3
		(iii)				
					2A	
						2
	(b)	(i)			1A	
			The angle of incidence of the light ray from the object is (45° , which is) less than the critical angle of the plastic prism.		1A	
			Total internal reflection will not occur and the image may not be clear enough for observation.		1A	
						3
	(ii)	Glass prism (with critical angle less than 45°)			1A	
			Or			
			Plane mirror		1A	
						1

8.	(a)				
			1A	Correct symbols of light bulb, variable resistor and voltmeter	
			1A	Correct positions	
			1A	Correct positive terminal connection for the voltmeter	
					3
	(b)	As the voltage across the light bulb increases, the temperature of the light bulb increases, thus its resistance increases.	1A		
			1A		
					2
	(c)	$R = \frac{V}{I}$ is the definition of resistance. It is applicable to all conductors.	1A		
					1
	(d)	At $V = 0.1$ V $R = \frac{V}{I} = \frac{0.1}{76 \times 10^{-3}} = 1.32 \Omega$ At $V = 2.5$ V $R = \frac{V}{I} = \frac{2.5}{250 \times 10^{-3}} = 10 \Omega$			
			1A		
			1A	for reading correct values (ignore the order of magnitude) from the graph	
			1A		
					3
	(e)	$R = \rho \frac{l}{A}$ $l = \frac{RA}{\rho}$ $= \frac{1.32 \times (1.66 \times 10^{-9})}{5.6 \times 10^{-8}}$ $= 0.039$ m			
			1M+1M		
			1A		
					3

7.	(a)	(i)	Angle of incidence at A , $i_A = 90^\circ - 30^\circ = 60^\circ$	1A	
				1	
		(ii)	$n_g \sin c = n_c \sin 90^\circ$ $\Rightarrow \frac{n_g}{n_c} = \frac{1}{\sin c} \geq \frac{1}{\sin 60^\circ} = 1.1547005 \approx 1.15$	1M	
				1A	
				2	
	(b)	(iii)	Total internal reflection. For light rays from O with $\theta > 30^\circ$, they will fail to undergo total internal reflection.	1A 1A	Correct spelling Accept $\theta \geq 30^\circ$
				2	
		(i)	Some light / energy (of the narrow light pulse) taking the <u>shortest path</u> (i.e. OD) <u>arrives first</u> while the rest of the energy taking <u>longer paths</u> <u>arrives later</u> . Therefore the pulse is broader and the height of the pulse is lower (smaller intensity).	1A	1A for 'taking different paths' 1A for 'different arrival time' NOT accept: Some of the energy (going at angles larger than 30° from the axis) would be lost due to leakage, thus lower intensity.
				1A	
				2	
		(ii)	The refractive index of cladding should be increased, as $\frac{n_g}{n_c} = \frac{1}{\sin c}$, by <u>making $c / \sin c$ larger</u> , only those light rays close to the axis / within a smaller θ are totally internal reflected, (such that $\frac{n_g}{n_c}$ getting closer to 1).	1A	0 A for the explanation if the choice for the change of n_c is INCORRECT. Note: More light rays will escape the optical fibre for a larger n_c .
				1A	
				2	

Solution	Marks	Remarks
6. (a) (i) $\sin r = \frac{\sin 45^\circ}{1.325}$ $r = 32.253454^\circ \approx 32.3^\circ$ (ii) $c = \sin^{-1} \frac{1}{1.325}$ $c = 49.000646^\circ \approx 49.0^\circ$ r (or $\angle OBA$) $= 32.3^\circ < c = 49.0^\circ$, therefore total internal reflection does not occur at point B.	1M 1A 2 1M 1A 1A 3	
(b) (i)  (ii) There is more light energy absorbed by water droplet due to a longer path. Or There is more light energy transmitted to air at the point of reflection as the light undergoes reflection twice.	2A 2 1A 1A 1A 1A 2	

8. (a)



(b)

$$1.33 = \frac{\sin r}{\sin 1.5^\circ}$$

$$r = 1.995175^\circ \approx 2.00^\circ$$

(c)

$$\frac{h_1}{h} = \frac{\tan 1.5^\circ}{\tan 2.00^\circ} = 0.751681 \text{ (or } \tan 1.5^\circ = \frac{PQ}{0.10} \text{ \& } \tan 2^\circ = \frac{PQ}{h_1})$$

$$\Rightarrow h_1 \approx 0.075 \text{ m (i.e. 7.5 cm)}$$

1A

1M

1M for image I formed by the two refracted rays.

2

1M

1A

2

1M

1A

e.c.f. from (b), explicit step / method shown

Accept: $\frac{\text{real depth}}{\text{apparent depth}} = \frac{h}{h_1} = n$

2